CS460 Assignment 3: Stereo Vision

Overview

Assignment 3 focuses on：

Conducting camera calibration and fundamental matrix computation.

Note

The knowledge required for Assignment 3 to complete the assignment is covered in detail in **preliminaries.pdf.** Please read **preliminaries.pdf** carefully before starting the assignment.

Technical Requirements

1. **Python Version** Ensure you are using **Python version 3.7 or higher**.
2. **Required Libraries** You will need to install the following Python libraries to complete the assignment. If you are unfamiliar with installing Python libraries, refer to online resources (e.g., using pip). The required libraries are:
    - **NumPy**: For numerical operations and array manipulation.
    - **Matplotlib**: For creating visualizations, such as plots and images.
    - **OpenCV (cv2)**: For image processing and visualization tasks. Please refer to the tutorial about how to install OpenCV for python https://www.runoob.com/opencv/opencv-install.html

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Assignment Tasks and Submission Requirements

Please refer to the **camera_and_stereo.pdf** document for a detailed description of the tasks and requirements. You are required to complete the following tasks:

- **Task1 and Task2 in camera_and_stereo.pdf**

The tasks highlighted in red in the document indicate the questions you need to answer in the code.

The tasks highlighted in blue in the document indicate the questions you need to discuss in the report.

Submission Format

1. Code file (Only the supplementary code file needs to be uploaded)
2. PDF document (The PDF document should elaborate on the questions to be discussed)

3. **Package the code file and PDF document into a compressed file for upload.** The naming format should be: Chinese Name + Student ID.
4. Please use the provided **cover_template.docx** as the report cover template.
5. Organize your PDF file logically, with clear section headers for each task. (e.g., "Task 1", "Task 2").